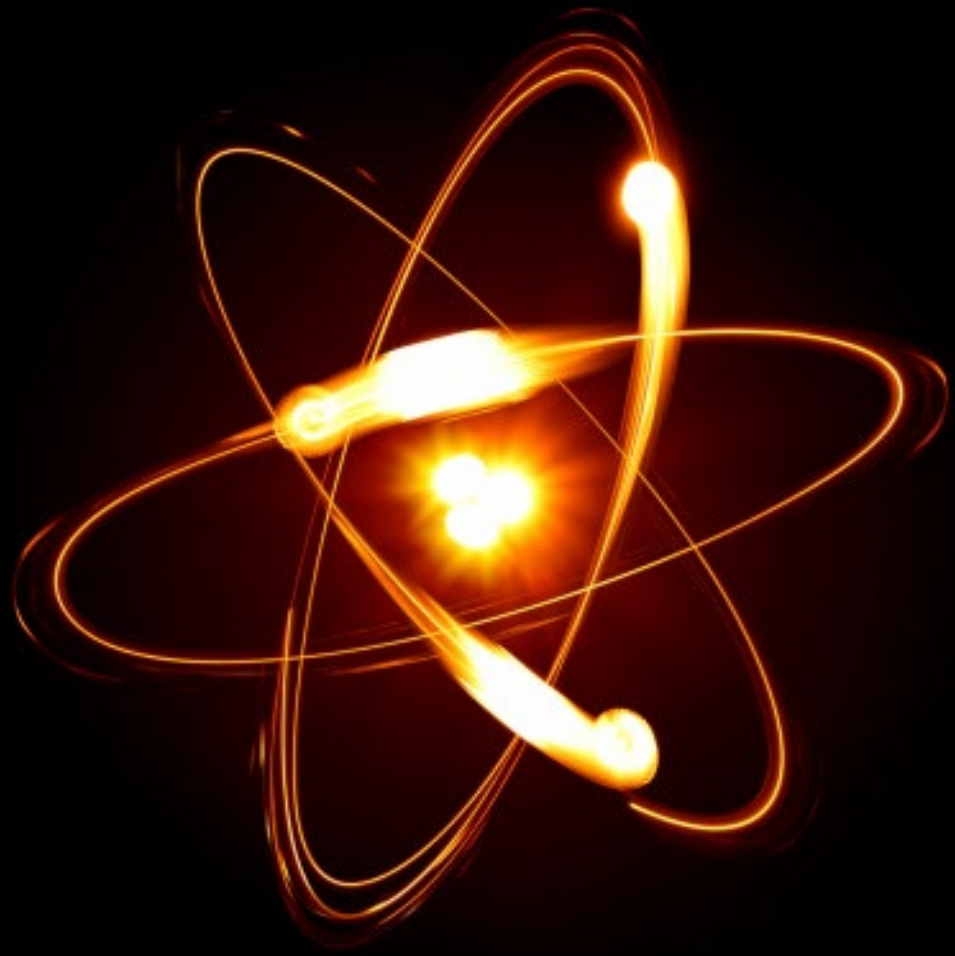
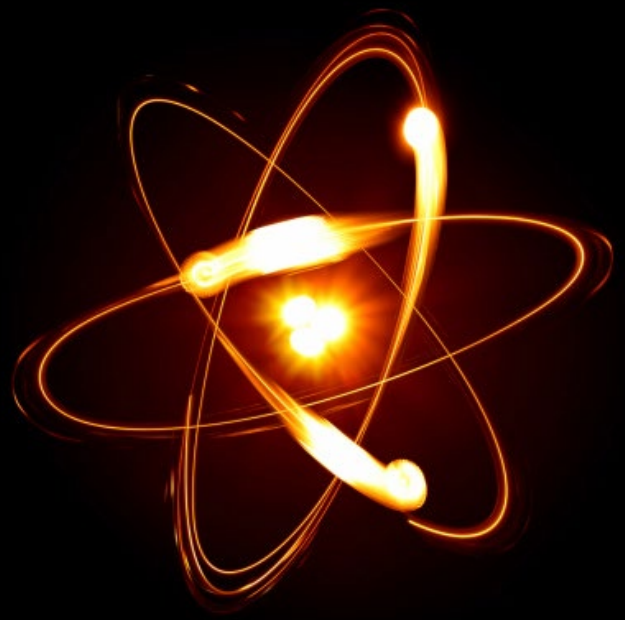
Chapter 2.

Photons. Particle-like properties of radiation



Beatriz González del Río

Quantum Physics



Chapter 2.

Photons. Particle-like properties of radiation

- Introduction
- The photoelectric effect
- Photoelectric effect: Einstein´s quantum interpretation
- Electromagnetic spectrum
- On the special theory of relativity. Photons
- The Compton effect
- X ray production
- Pair production and pair annihilation

Introduction

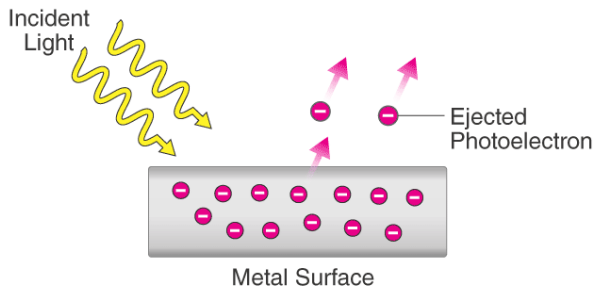
Interaction of radiation with matter.

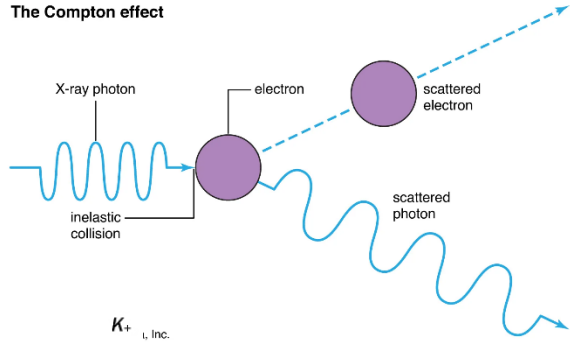
Examine five different phenomena:

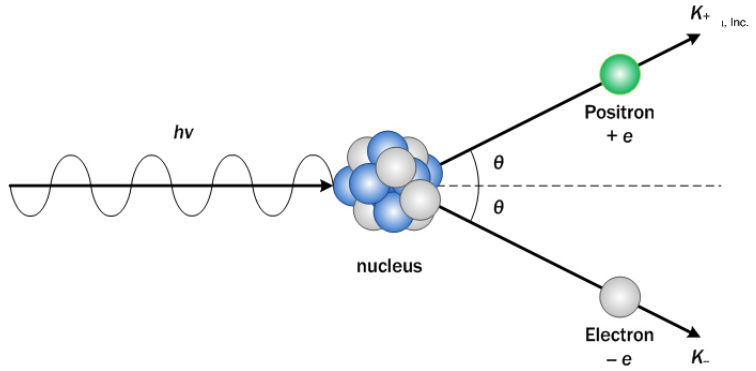
1) **Absortion or Scattering of radiation**
   - Photoelectric Effect
   - Compton Effect
   - Pair Production

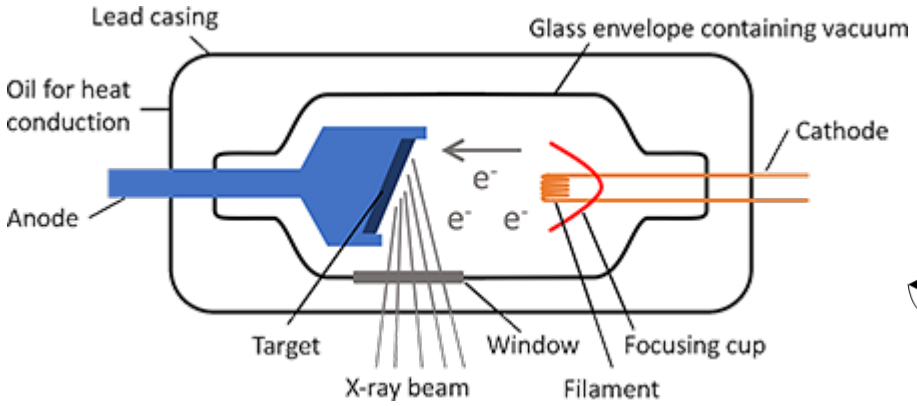
2) **Production of radiation**
   - X-ray production
   - Pair Annihilation

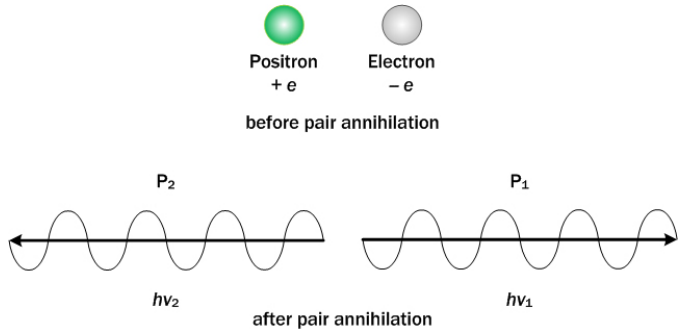
When radiation interacts with matter, it behaves in a particle-like way.











The Photoelectric Effect

- Discovered by Hertz in 1889
- Lenard (1899) showed that ultraviolet light makes electrons to be ejected from the cathode.

Experimental setup:

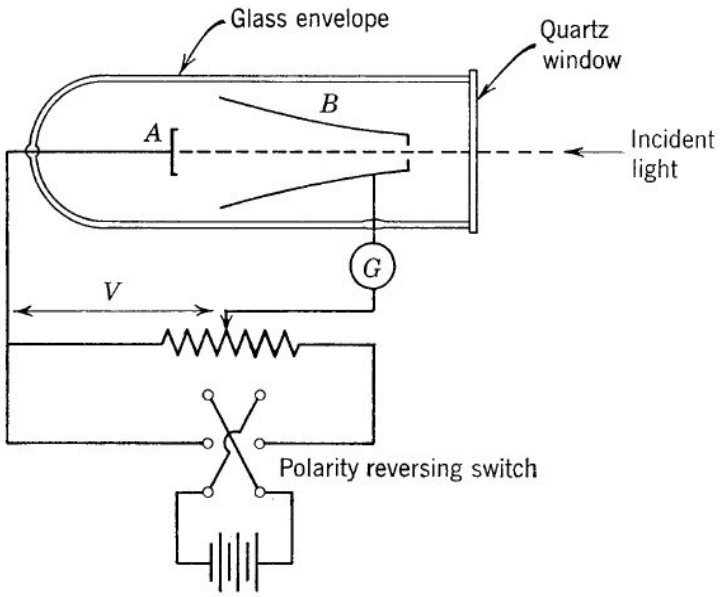


Figure 2-1 An apparatus used to study the photoelectric effect. The potential difference V can be varied continuously in magnitude, and also reversed in sign by the switching arrangement.

When light (monocromatic light) falls on the metal plate A, then an electric current appears in the circuit.

The Photoelectric Effect. Experimental properties

Fotoelectric current vs potential

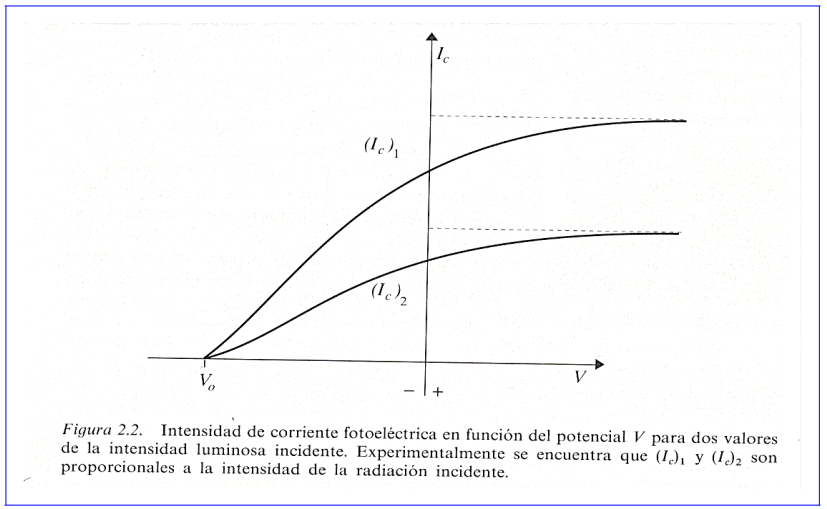
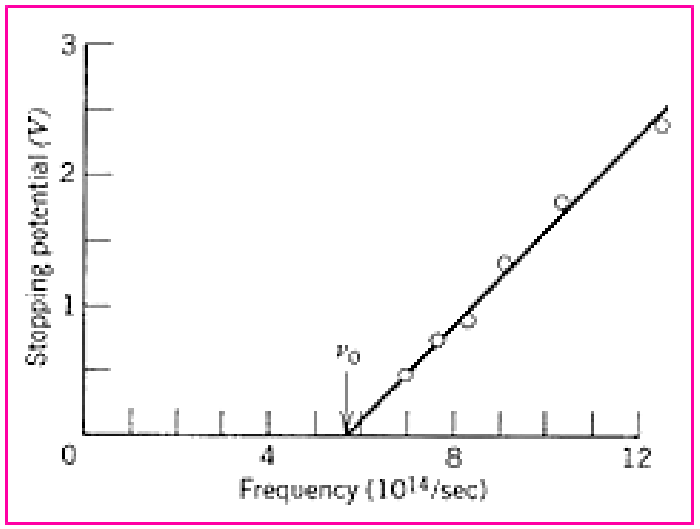


Figura 2.2. Intensidad de corriente fotoeléctrica en función del potencial V para dos valores de la intensidad luminosa incidente. Experimentalmente se encuentra que $(I_c)_1$ y $(I_c)_2$ son proporcionales a la intensidad de la radiación incidente.



Properties

- The number of ejected electrons is proportional to the light intensity.
- When V is large enough, the photoelectric current reaches a certain limit.
- There is a stopping potential, V_0, at which the photoelectric current drops to zero.

$$e|V_0| = K_{max}$$

- K_{max} is independent of the light intensity.
- Not any light can produce a photoelectric effect. There is a minimum frequency ν_0, (cutoff frequency) below which there is NO effect.

The Photoelectric Effect. Classical interpretation

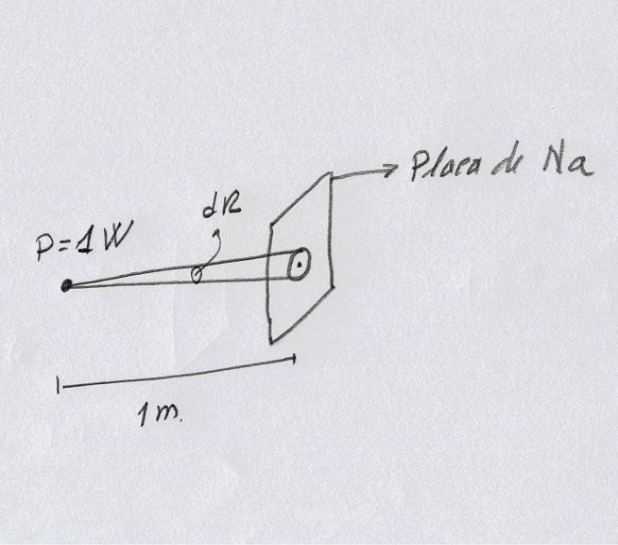
According to classical EM theory, what should happen if we increase the intensity?

Classical interpretation: The electromagnetic waves induce an oscillating behavior on the electrons. When the amplitude becomes large enough, the electron is ejected. But:

- The electron feels a force (-e)E. Increasing the intensity means increasing E and therefore the force on the electron. Consequently the $K_{max} = e|V_0|$ should increase with the light intensity.
- There should be photoelectric effect for any frequency as long as it is intense enough.
- There should be a time lag because in the classical theory the light energy is uniformly distributed over the wave front; however no detectable time lag has ever been measured.

Time lag calculation

Consider a light source whose power is 1 W = 1 J/s and there is a Na plate placed at 1 m.



The ejected photoelectron collects its energy from a circular area on the plate, whose radius is ≈ 1 Å.

Total area for collecting energy is = $\pi r^2 = \pi\ 10^{-20}$ m^2

The energy from the source reaches this area at a rate R

$$R = 1\frac{J}{s}\ \frac{1}{4\pi 1m^2}\pi 10^{-20}m^2 = 2.5\ 10^{-21}\ J/s$$

The energy needed to eject a photoelectron is 2.1 eV = 3.4 10^{-19} joules. Then, it would take a time

$$t = \frac{3.4\ 10^{-19}\ J}{2.5\ 10^{-21}\ J/s} = 2\ minutes$$

In 1928, Lawrence and Beams measured the time lag and found that it should be less than $3{\cdot}10^{-9}$ s.

The Photoelectric Effect. Einstein's Quantum Interpretation

Einstein (1905). New theory of the electromagnetic radiation

- Radiant energy is quantized into concentrated bundles (**photons**)
- Each photon has an energy $\mathbf{E} = \boldsymbol{h\nu}$ and in the photoelectric process one photon is completely absorbed by one electron.
- The maximum energy of an ejected photoelectron is $\boldsymbol{K_{max} = h\nu - \omega_0}$, where ω_0 is the work function (minimum energy required to escape from the metal).

TABLA 2.1.

Funciones de trabajo para diversos metales.

Metal		Función de trabajo (eV)	Metal		Función de trabajo (eV)
Litio	Li	2.4	Aluminio	Al	4.2
Sodio	Na	2.1	Wolframio	W	4.5
Potasio	K	2.0	Platino	Pt	6.3

The Photoelectric Effect. Einstein's Quantum Interpretation

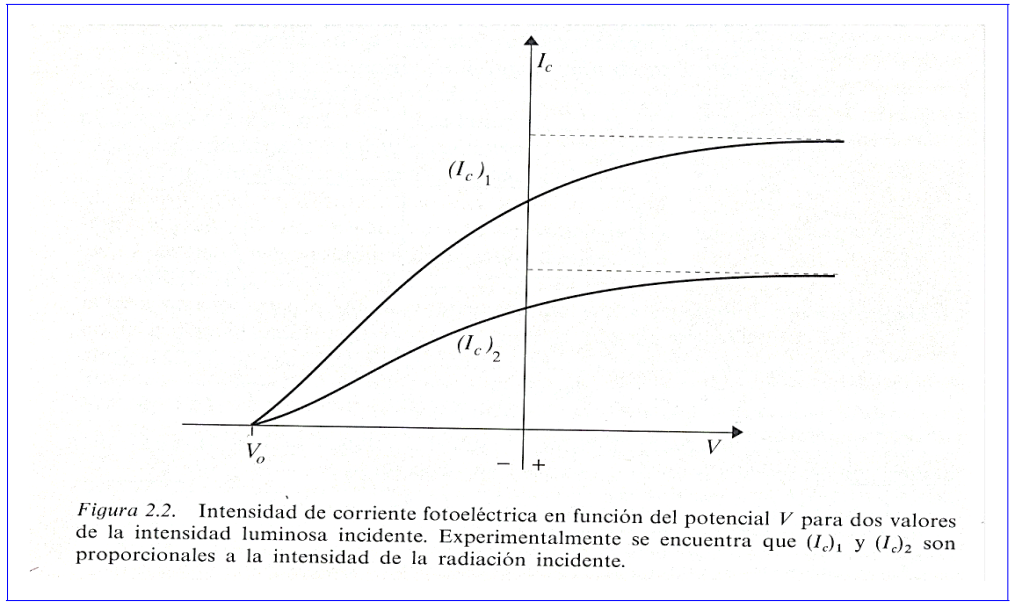
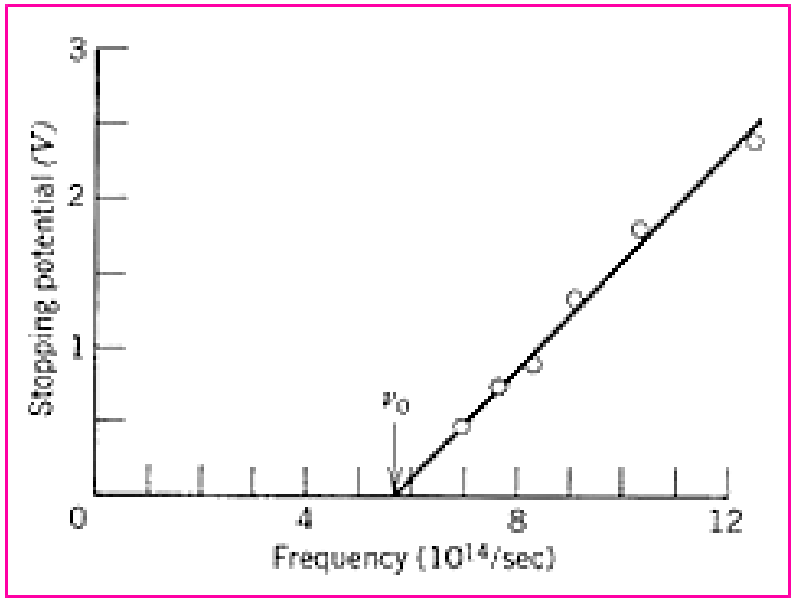


Figura 2.2. Intensidad de corriente fotoeléctrica en función del potencial V para dos valores de la intensidad luminosa incidente. Experimentalmente se encuentra que $(I_c)_1$ y $(I_c)_2$ son proporcionales a la intensidad de la radiación incidente.



When the electron is emitted, its kinetic energy will be : $K = h\nu - \omega$

When ω takes its minimum value $= \omega_0$, then $K = K_{max}$.

Now, $e|V_0| =$ energy of the most energetic electrons $\equiv K_{max}$

What do the slope and the x-intercept represent?

$$e|V_0| = K_{max} = h\nu - \omega_0$$

The Photoelectric Effect. Einstein's Quantum Interpretation

Explanation of the previous objections:

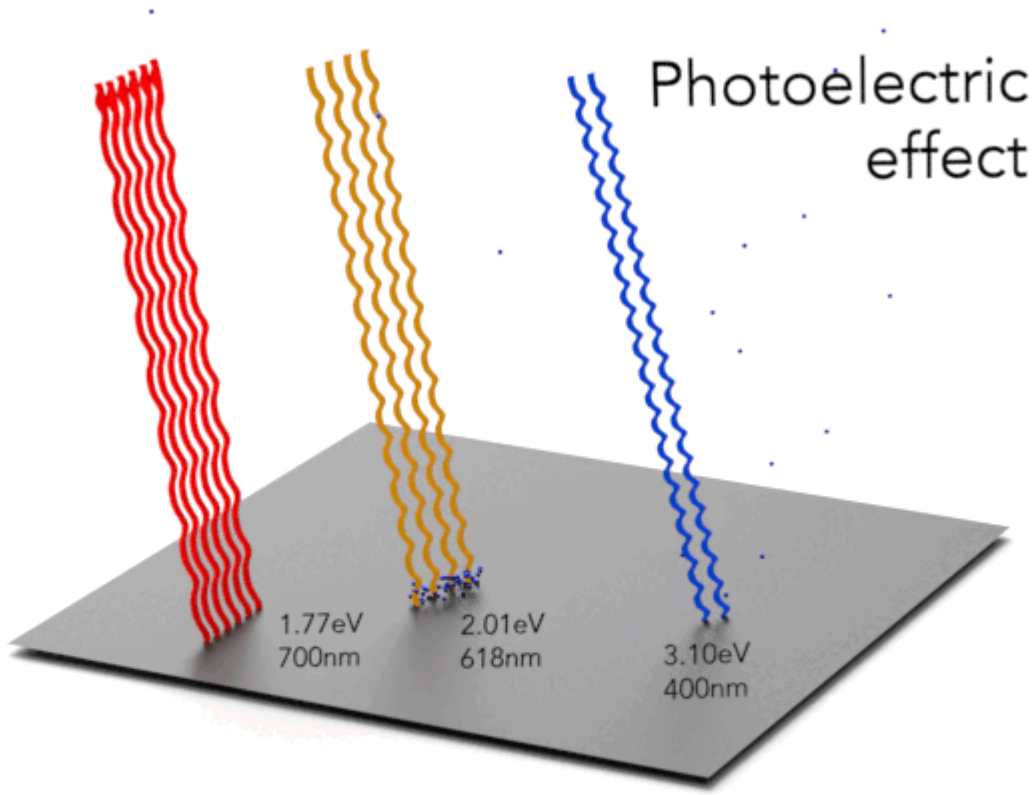
- No relation between K_{max} and light intensity ?

  Increasing light intensity means increasing the number of photons BUT the energy of the photons remains the same.

- Existence of a cutoff frequency, ν_0 ?

  It appears when $K_{max} = 0$ because now $0 = h\nu_0 - \omega_0$. Therefore $h\nu_0 = \omega_0$.

- No time lag



Potassium - 2eV needed to eject electron

The Photoelectric Effect. Einstein's Quantum Interpretation

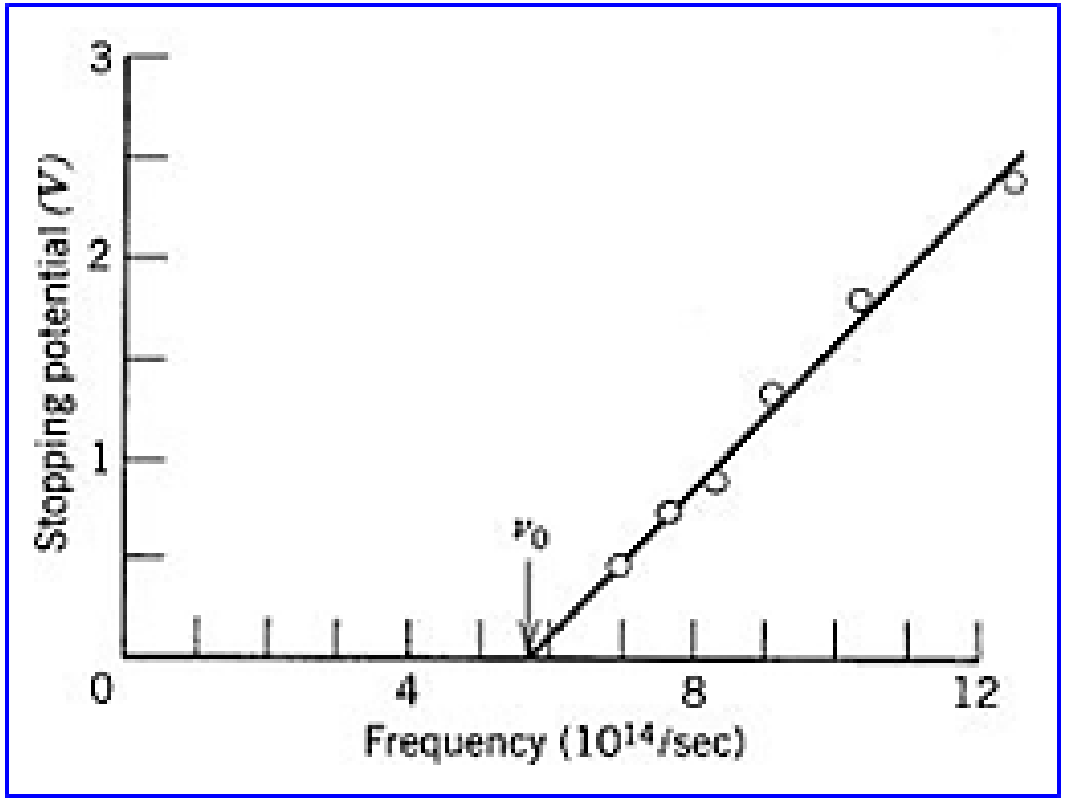
Write Einstein's eqn.

$$e|V_0| = K_{max} = h\nu - \omega_0$$

What should you plot to determine h ?

$$|V_0| = \frac{h}{e}\nu - \frac{\omega_0}{e}$$

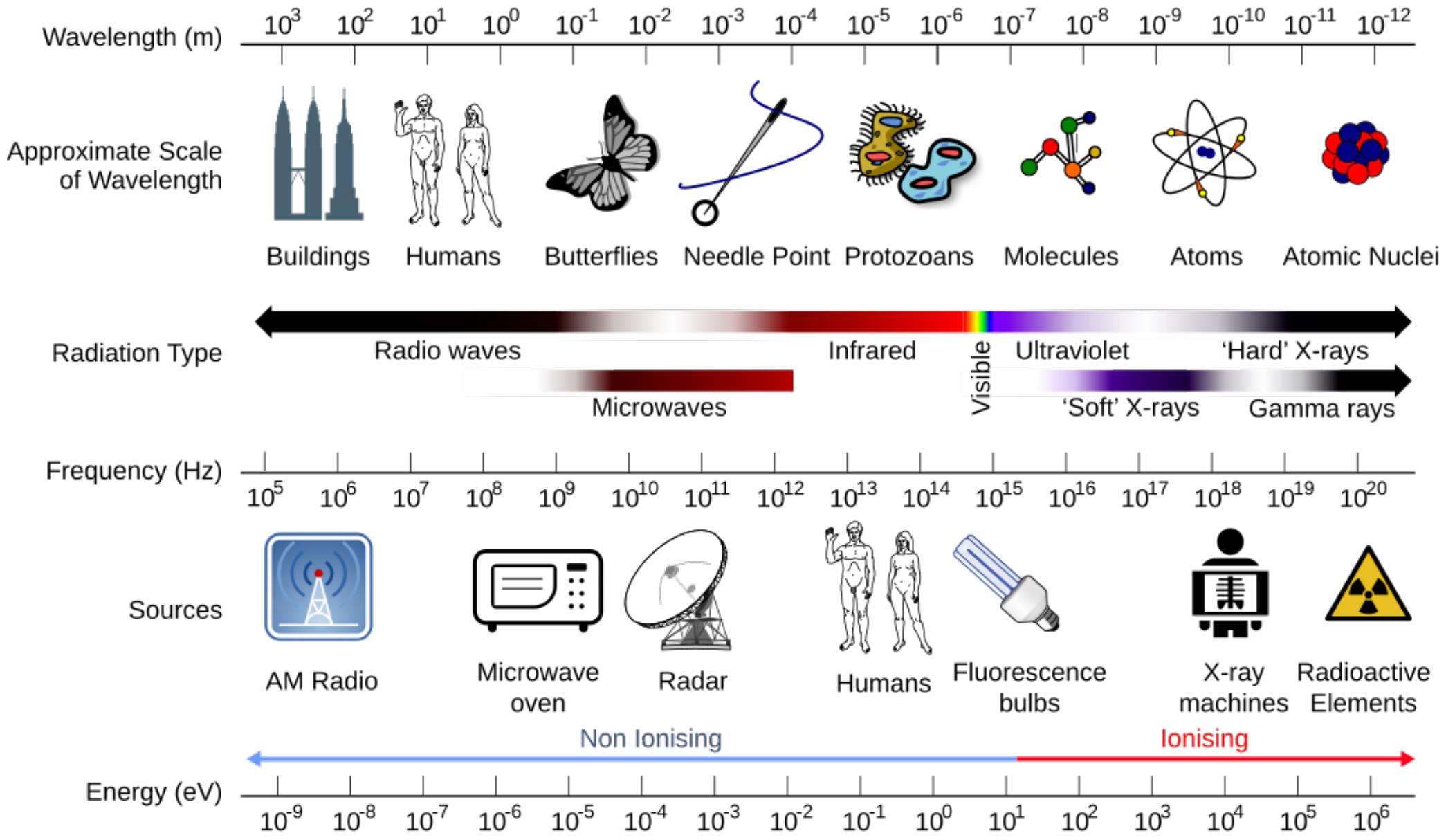
Relation between the stopping potential and the frequency



It can be used to evaluate h

YOU WILL DO EXACTLY THIS IN THE LAB

Electromagnetic spectrum



On the special theory of relativity. Particles

Basic equations for a free body

$$E^2 = p^2c^2 + m^2c^4 \qquad \vec{p} = \vec{v}\frac{E}{c^2}$$

E = total energy

$\vec{p}$ = linear momentum

$\vec{v}$ = velocity

m = mass (same as in classical mechanics)

Suppose $\vec{v} = 0 \Rightarrow \vec{p} = 0 \Rightarrow E^2 = m^2c^4 \Rightarrow E = mc^2 \equiv E_0$, where E_0 is the rest energy

Assuming that $v \ll c \Longrightarrow \frac{v^2}{c^2} \approx 0$, then

$$E^2 \approx m^2c^4 = E_0$$

In general

$$\mathrm{E} = E_0 + K = \sqrt{p^2c^2 + m^2c^4} = mc^2\sqrt{1 + \frac{p^2}{m^2c^2}} = \ \ldots = mc^2 + \frac{p^2}{2m} - \frac{1}{8}\frac{p^4}{m^3c^4} + \cdots$$

On the special theory of relativity. Photons

Apply the previous eqns. to the photons.

According to Einstein, the photons move with a velocity $v \equiv c$

Then (second eqn.) $E = pc$

Now,

$$E^2 - p^2c^2 = m^2c^4 \Longrightarrow p^2c^2 - p^2c^2 = m^2c^4 = 0 \Longrightarrow m = 0 \Longrightarrow E_0 = 0$$

Its energy is only kinetic energy

On the other hand

$$\mathrm{E} = \mathrm{pc} = \mathrm{h}\nu \Longrightarrow \mathrm{p} = \frac{\mathrm{h}\nu}{\mathrm{c}} = \frac{\mathrm{h}}{\lambda}$$

Connects a particle-like feature (p) with a wave-like magnitude (λ)

The Compton Effect

This experiment highlights the corpuscular (particle-like) nature of radiation. (A.H. Compton, 1923)

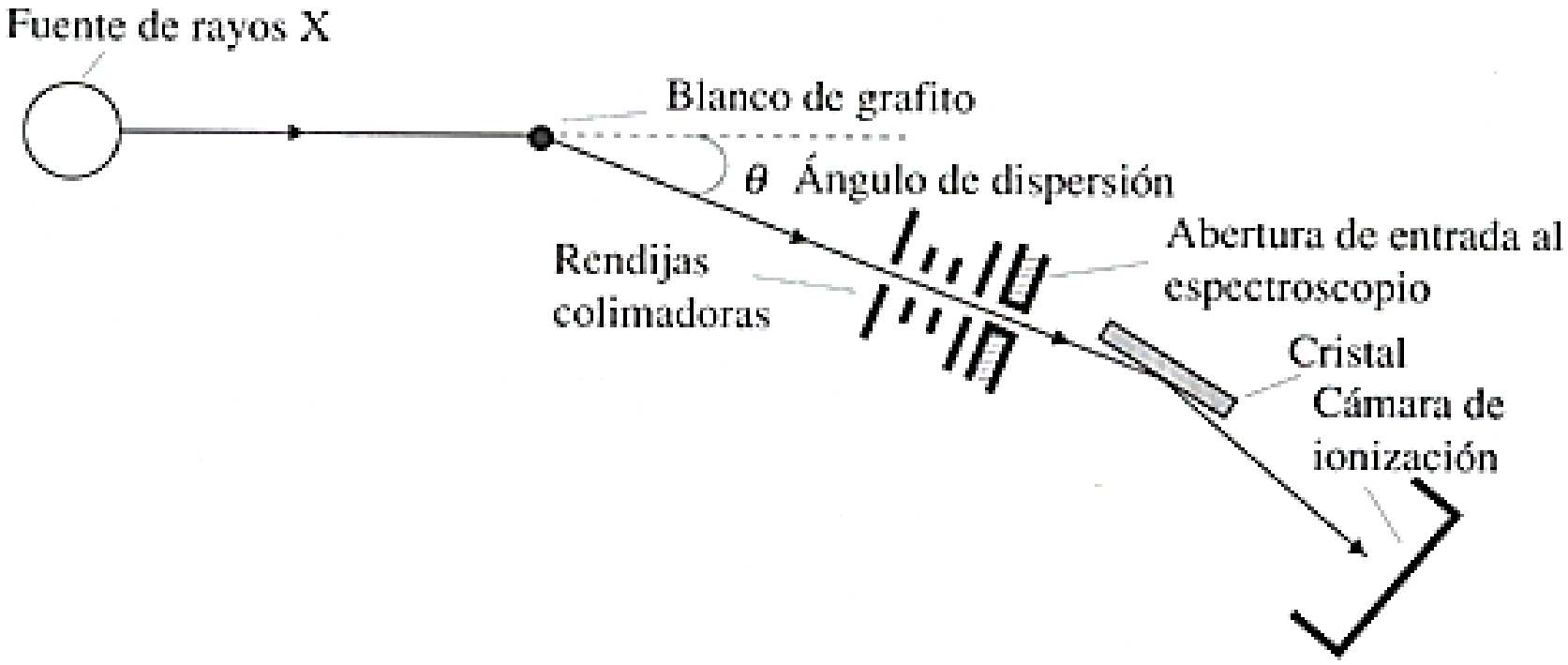
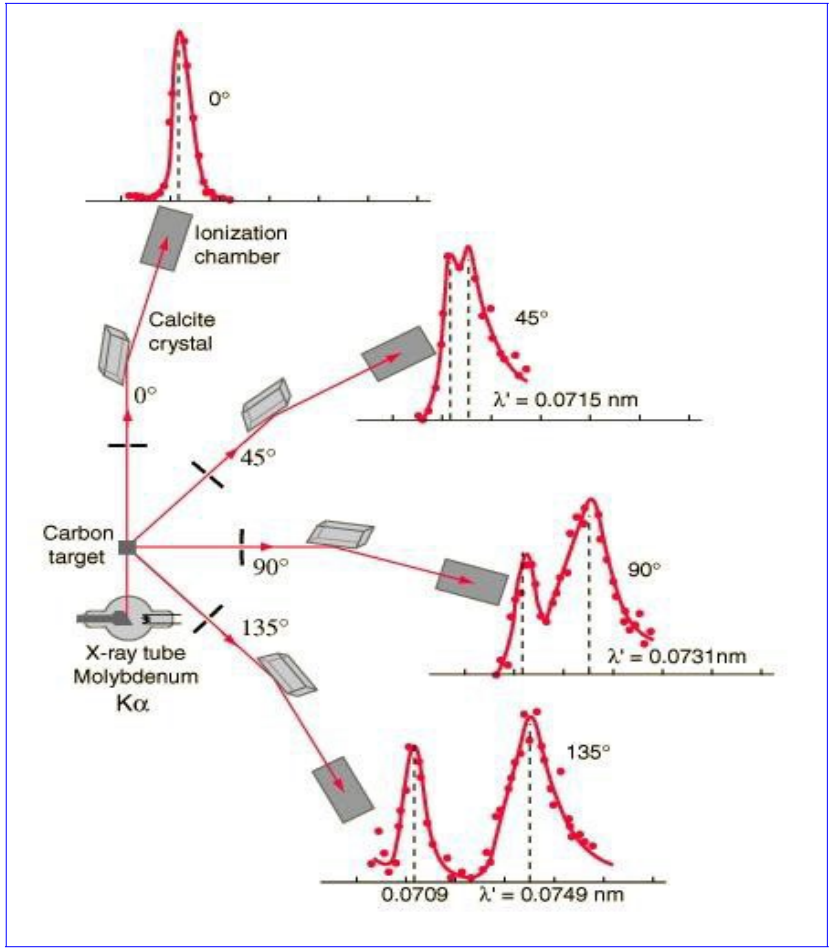


Figura 2.6. Esquema del montaje experimental utilizado por Compton.

The Compton Effect. Experimental data

Experimental data



Main results

- The scattered X rays have intensity peaks at TWO wavelengths.

  One is the same as the incident wavelength, λ_0, and the other, λ_1, is larger by an amount $\Delta\lambda = \lambda_1 - \lambda_0$

  This is clearly visible for scattering angles greater that 45^0.

- $\Delta\lambda$ depends on θ only. Moreover, it increases with θ.

- As θ increases, the intensity of the peak at λ_0 decreases whereas that of the peak at λ_1, increases.

The Compton Effect. Different targets

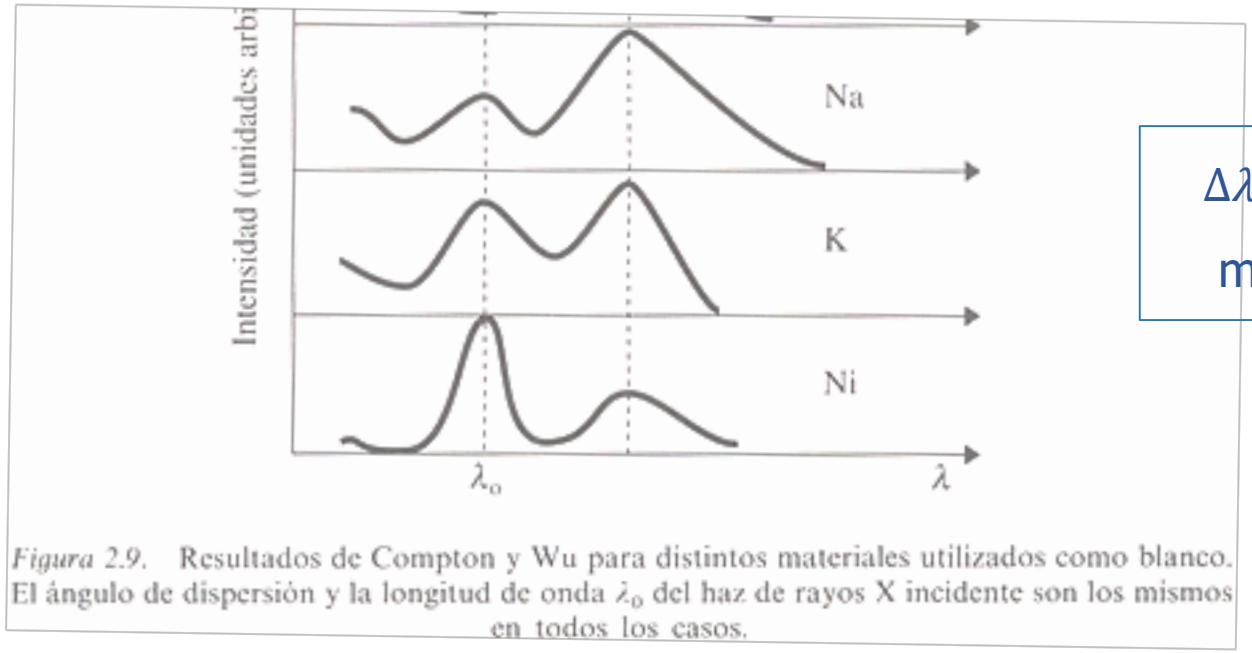


Figura 2.9. Resultados de Compton y Wu para distintos materiales utilizados como blanco. El ángulo de dispersión y la longitud de onda λ_o del haz de rayos X incidente son los mismos en todos los casos.

$\Delta\lambda$ is independent of the material used as target

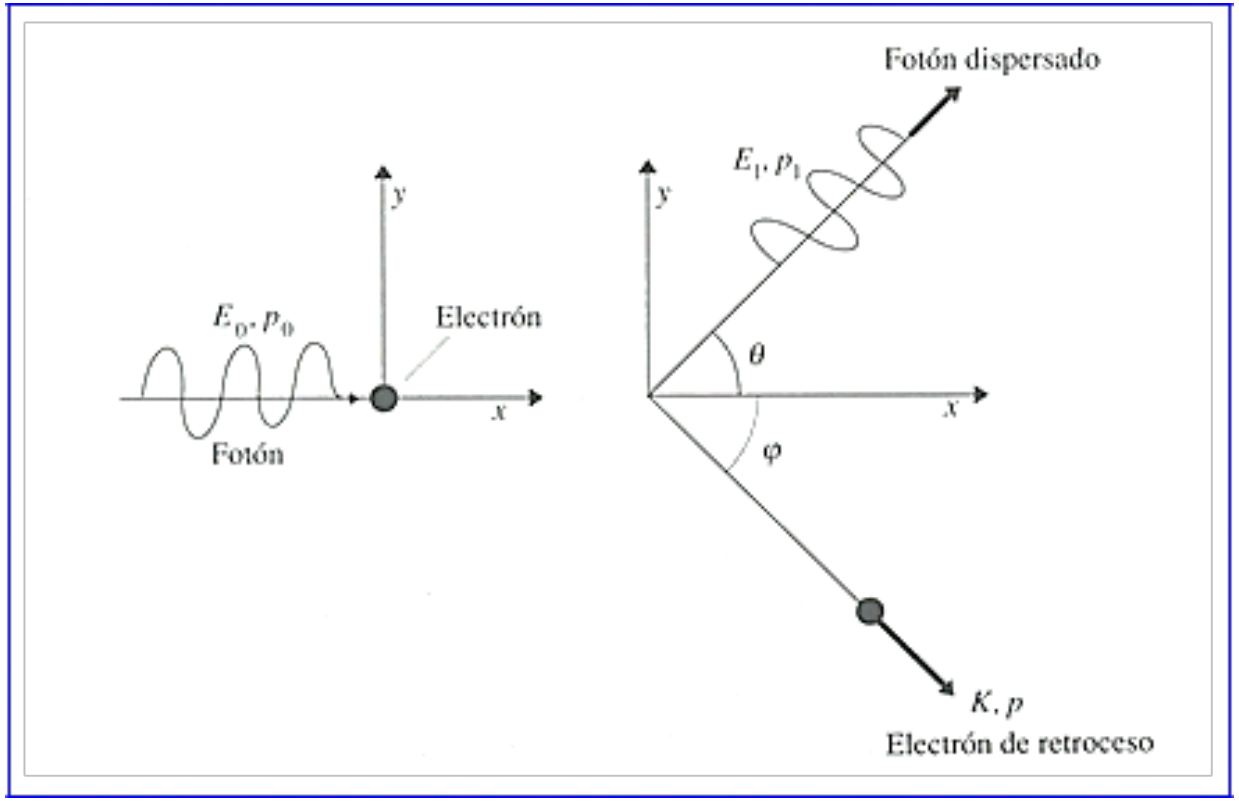
What does this suggest about what the photon is actually colliding with?

The whole atoms play no role in this process

The intensity of the peak at λ_0 increases with the atomic number (Z) whereas that of the peak at λ_1 decreases.

The Compton Effect. Quantum interpretation

The interaction of the radiation with the electrons in the target can be explained as a collision between two free particles (a photon and an electron).



If the electron gains kinetic energy, should the scattered photon have a longer or shorter wavelength?

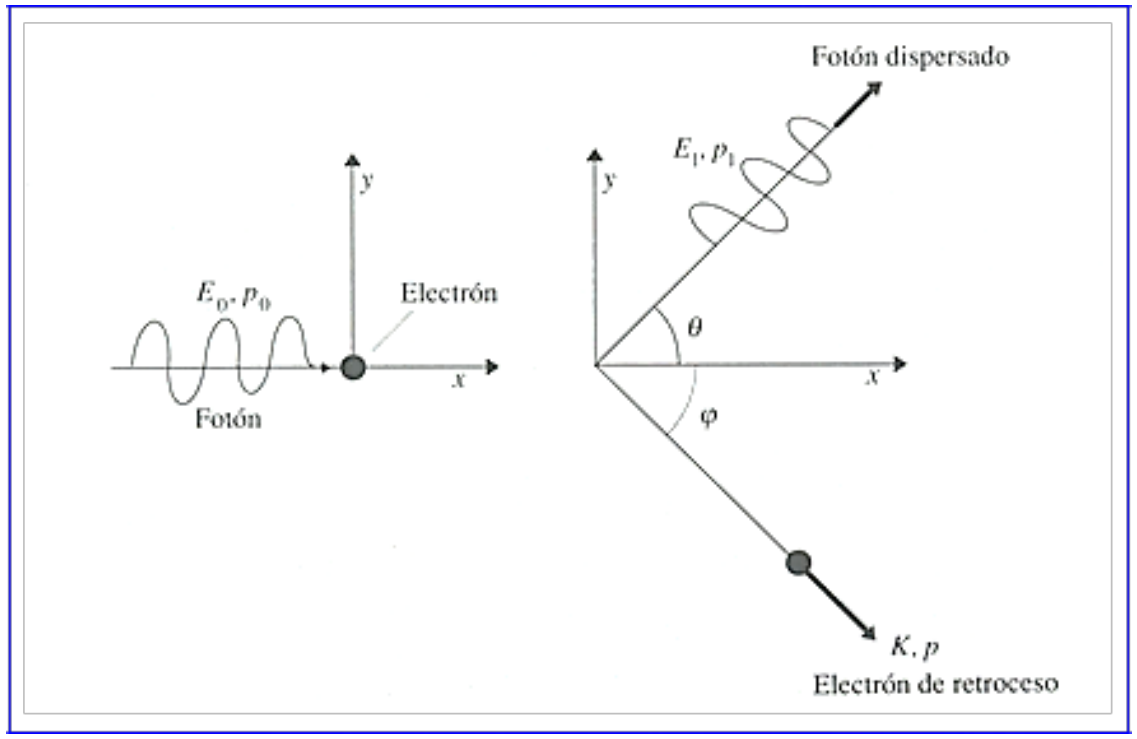
In the collision, the photon transfers some energy to the electron and becomes a photon with a lower frequency (or greater λ).

This explains the appearance of another maximum at greater λ.

The Compton Effect. Conservation laws

Assume that the electrons involved in the scattering process are free and initially stationary

Conservation of linear momentum and total relativistic energy.



p_0 is the momentum of the incident photon and p_1 that of the photon after the collision. p_e is the momentum of the electron after the collision.

Momentum conservation

$$p_0 = p_1 \cos\theta + p_e \cos\varphi \qquad \text{X-axis}$$

$$p_1 \sin\theta = p_e \sin\varphi \qquad \text{Y-axis}$$

Squaring these eqns. and adding them

$$p_e^2 = p_0^2 + p_1^2 - 2p_0 p_1 \cos\theta$$

The Compton Effect. Conservation laws

Conservation of total relativistic energy:

$$E_0 + m_e c^2 = E_1 + c\sqrt{p_e^2 + m_e^2 c^2}$$

Evaluating p_e

$$p_e^2 = \frac{(E_0 - E_1 + m_e c^2)^2}{c^2} - m_e^2 c^2$$

Inserting it into momentum conservation eq.

$$p_0^2 + p_1^2 - 2p_0 p_1 \cos\theta = \frac{(E_0 - E_1 + m_e c^2)^2}{c^2} - m_e^2 c^2$$

Taking into account that $E_0 = p_0 c$ y $E_1 = p_1 c$

$$p_0^2 + p_1^2 - 2p_0 p_1 \cos\theta = \frac{c^2(p_0 - p_1 + m_e c)^2}{c^2} - m_e^2 c^2$$

$$p_0^2 + p_1^2 - 2p_0 p_1 \cos\theta = p_0^2 + p_1^2 - 2p_0 p_1 + m_e^2 c^2 + 2m_e c(p_0 - p_1) - m_e^2 c^2$$

The Compton formula

$$p_0 p_1 (1 - \cos\theta) = (p_0 - p_1) m_e c$$

Multiplying by $\frac{h}{p_0 p_1 m_e c}$ and recalling that $\frac{h}{p_0} = \lambda_0$ and $\frac{h}{p_1} = \lambda_1$

$$\lambda_1 - \lambda_0 = \frac{h}{m_e c}(1 - \cos\theta) = \lambda_c (1 - \cos\theta)$$

where λ_c is the so-called Compton wavelength (0.02426 Å)

The Compton Effect. Explanation

The peak at λ_0 is due to a collision between a photon and an electron which is strongly bound to the atom. In fact, this amounts to a collision between the photon and the whole atom.

In this case $\lambda_c \approx 10^{-6}\text{Å}$, which explains the greater intensity of the peak at λ_0 when using as targets elements with a high Z (a greater number of internal electrons implies a greater probability of a collision with any of them).

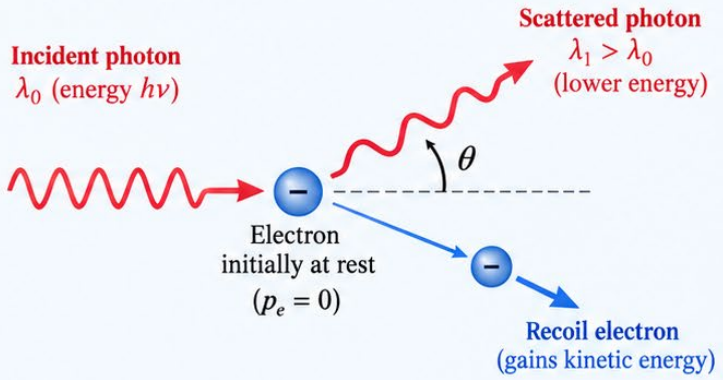
Some photons are scattered from electrons which are freed by the collision; then these photons are modified in wavelength.

Other photons are scattered from electrons which remain bound during the collision; then these photons are NOT modified in wavelength.

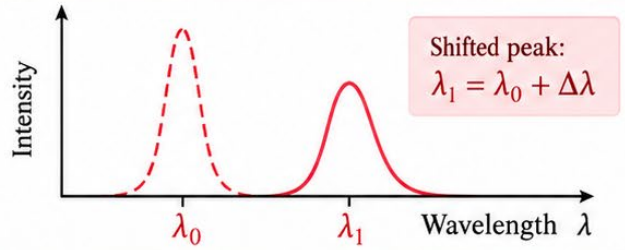
The Compton Effect.

Two collision cases in the Compton effect

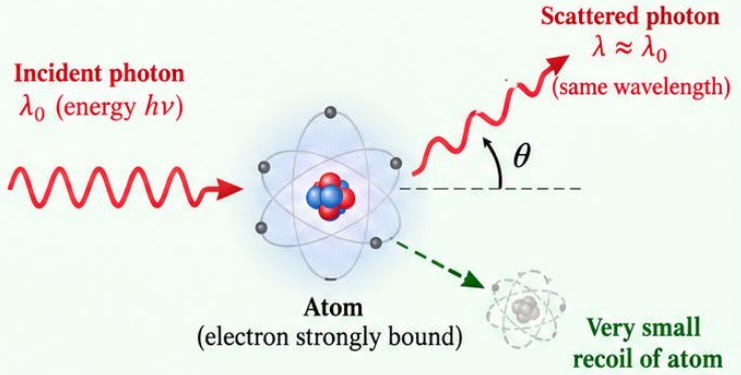
1. Collision with a quasi-free electron



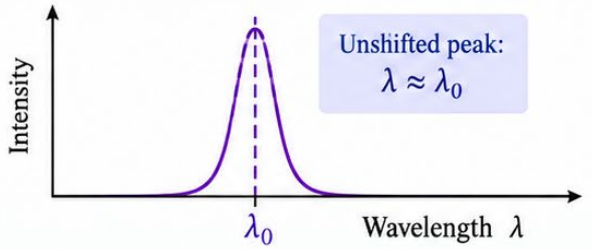
- Photon transfers energy and momentum to the electron.
- Scattered photon has lower energy and longer wavelength.



2. Collision with a strongly bound electron / whole atom



- The atom takes the recoil.
- Wavelength change is negligible.



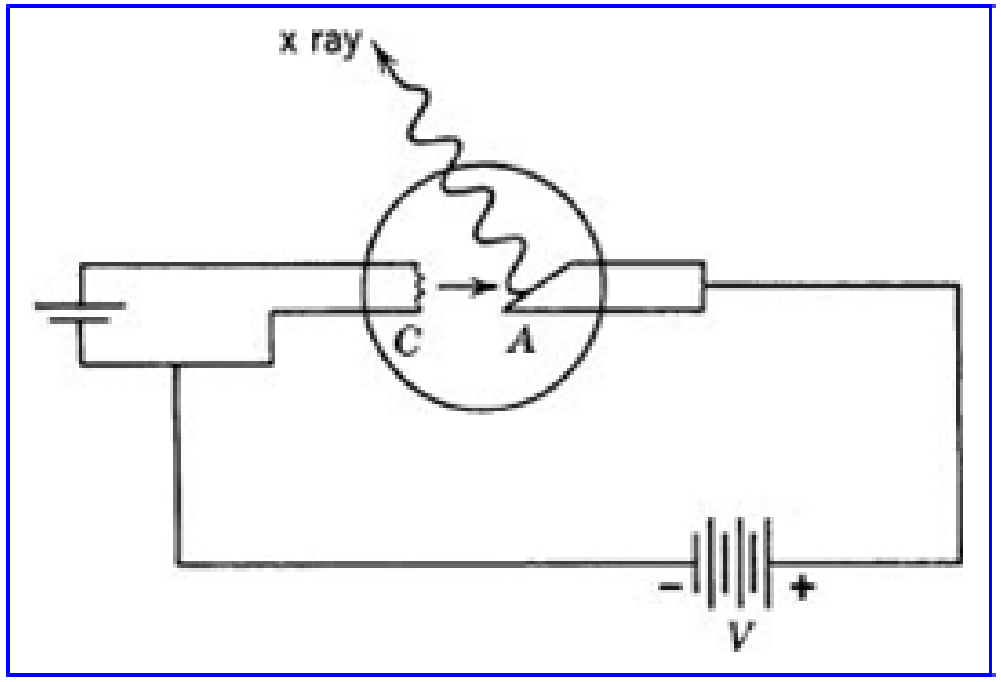
Observed spectrum:

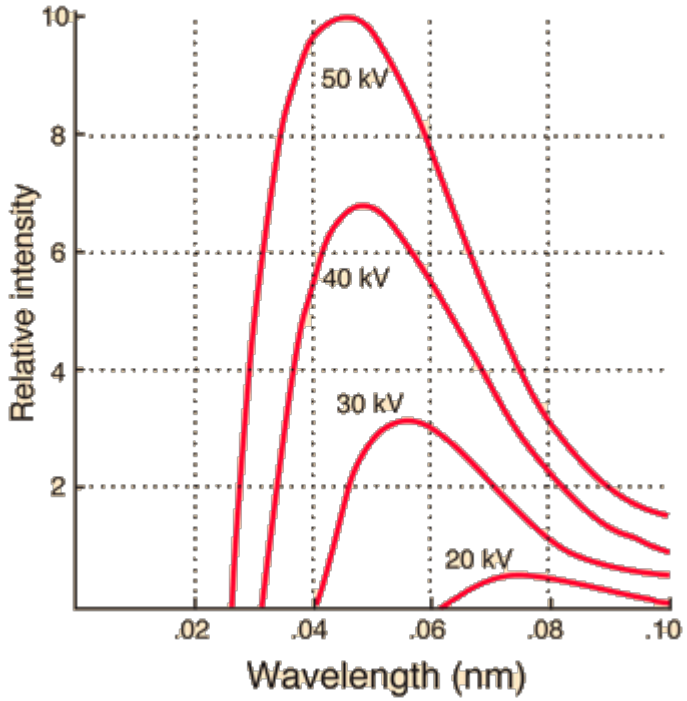
unshifted peak from bound-electron/atom scattering + **shifted peak** from quasi-free-electron scattering.

X ray production

X rays: Electromagnetic radiation whose wavelength is ≤ 1 Å.

A beam of electrons is accelerated through a potential difference of several thousand volts and it is directed against a target material. Then, we obtain a continuous spectrum of electromagnetic radiation.





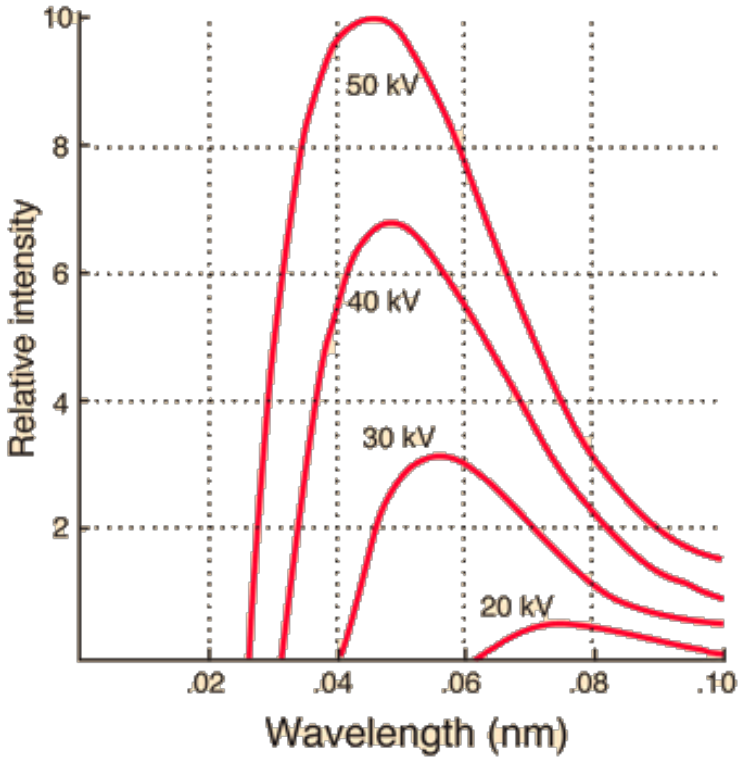
Continuous X-ray spectrum emitted by a W target, for different values of the potential difference.

For each curve exists a cutoff wavelength.

X ray production

Experimental data

X ray spectrum



Properties

- There is a λ_{min},

  e.g. λ_{min}= 0.311 for V=40 kV.

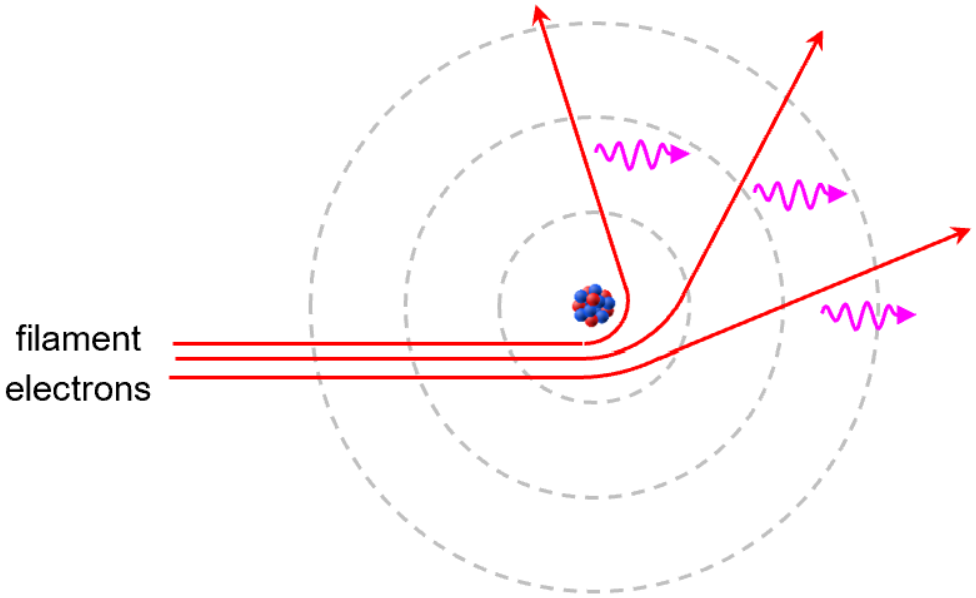
- The λ_{min} depends only on V.

- The λ_{min} is the same of all target materials

X ray production

Classical interpretation

Any electric charge which is accelerated (or decelerated) emits electromagnetic radiation. When an electron moves near a nucleus, it is decelerated and the energy lost by the electron is transformed into radiation.



An electron undergoes different decelerations (it interacts with many nuclei at different distances). This explains the appearance of a continuous spectrum.

NO explanation for the existence of a λ_{min}.

X ray production

Quantum interpretation

The different electrons move at different distances from the nuclei and therefore they undergo different decelerations.

In each encounter, the electrons lose some kinetic energy which is transformed into photons with a frequency:

$h\nu$ = lost energy = $K - K'$

(and this explains the continuous spectrum).

It may happen that an electron loses all its kinetic energy (eV) in a single interaction. This is transformed into a single photon whose energy is $h\nu_{max}$, i.e., $\nu_{max} = c/\lambda_{min}$. Therefore

$$\lambda_{min} = \frac{hc}{eV}$$

This equation allows to determine *h* from experimental data for the λ_{min}.

(note that if $h \longrightarrow 0$, then $\lambda_{min} \longrightarrow 0$ and we recover classical result)

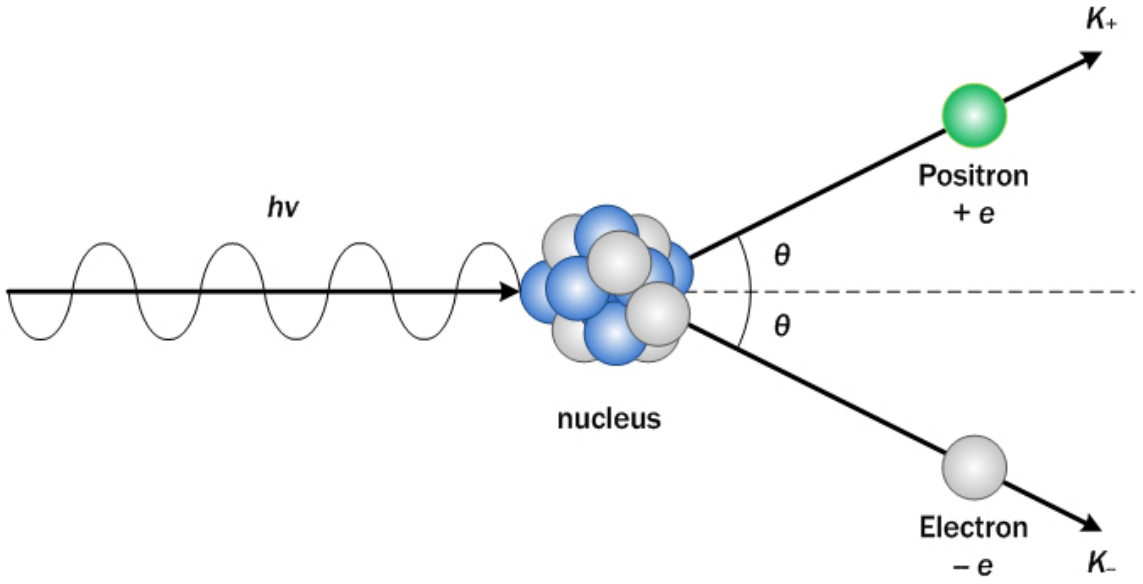
Pair production and Pair annihilation

Pair production: This is another process whereby photons lose their energy in interactions with matter.

In this process a photon loses all its energy (disappears) creating an electron and a positron (or proton and antiproton, ...).

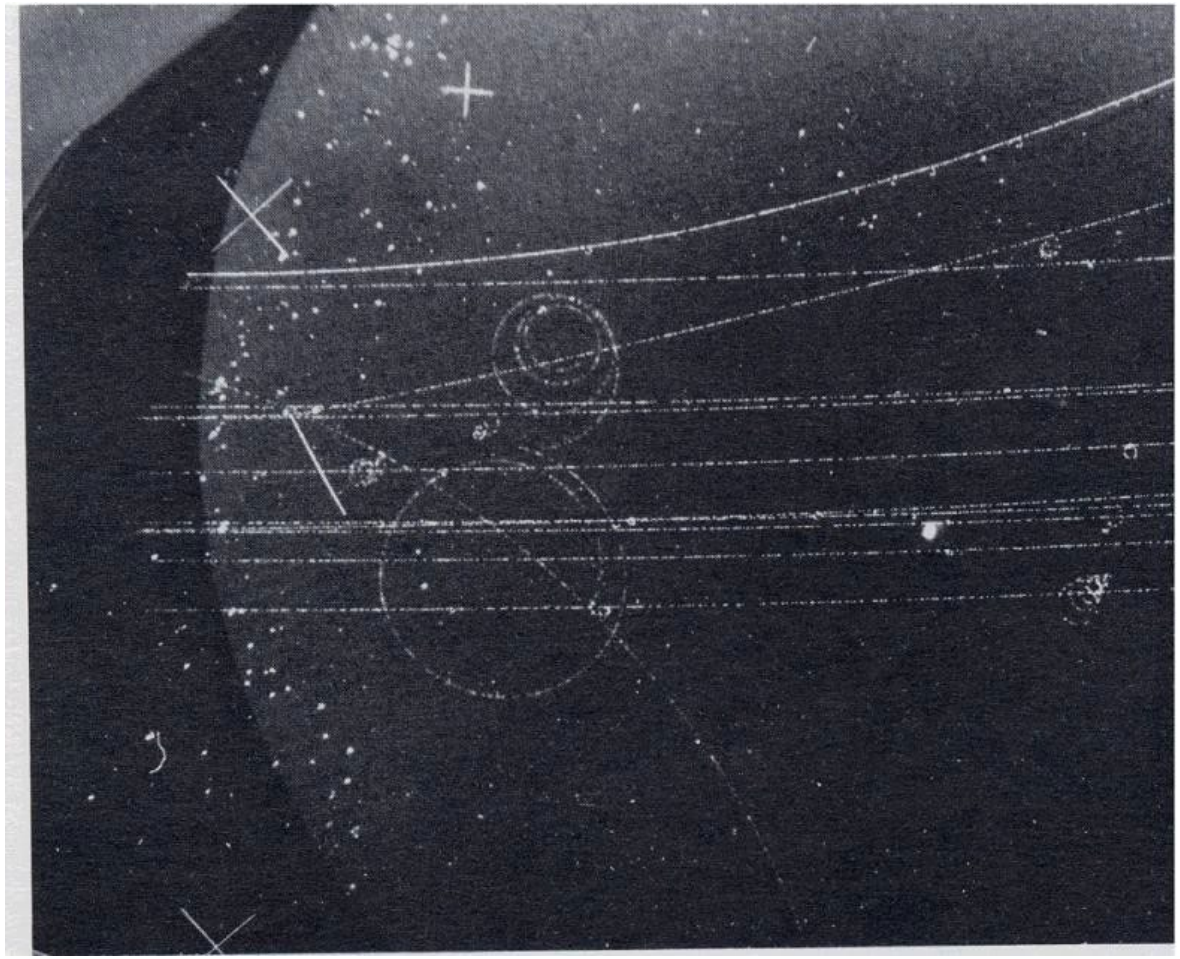
The first experimental evidence was provided by Anderson in 1933 while investigating the cosmic radiation.

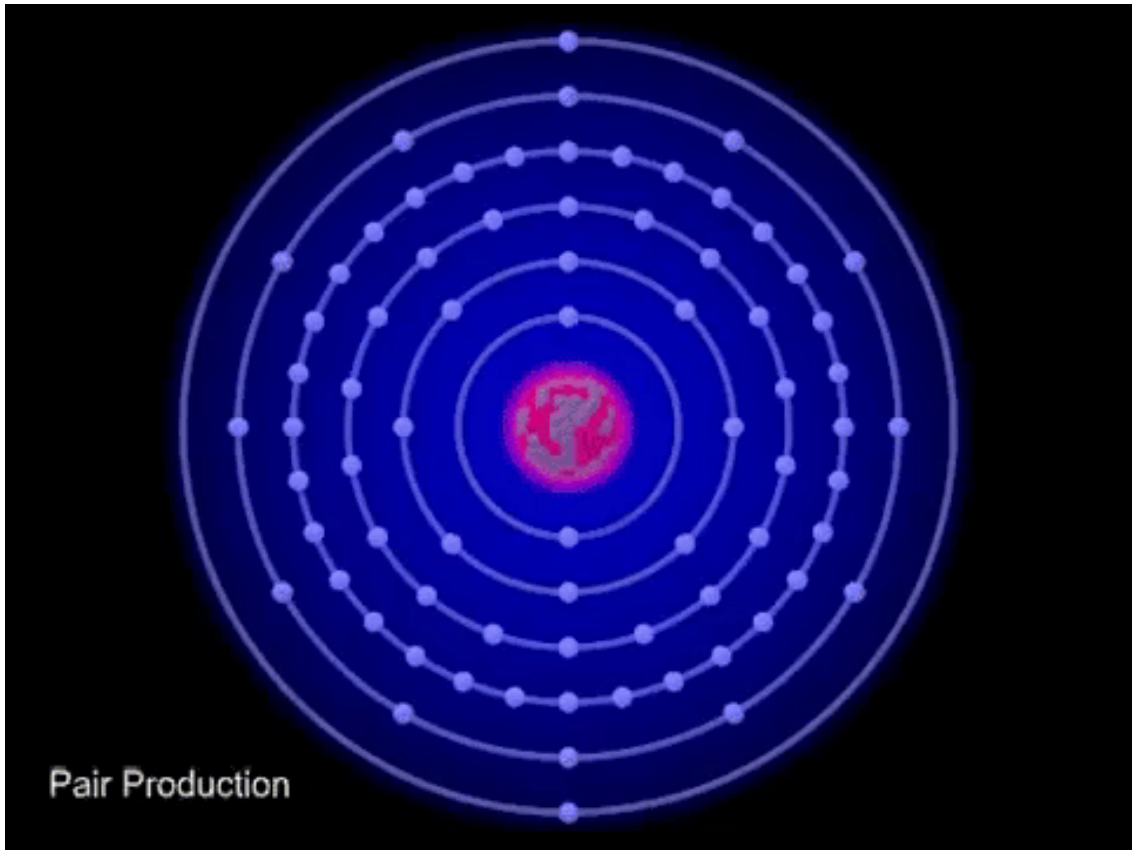
The process can be represented as conservation of total relativistic energy, momentum and charge



Pair production and Pair annihilation

Pair production as seen in a bubble chamber





Pair production and Pair annihilation

The calculation of the energies at play (energy conservation) goes as

$$h\nu = E_- + E_+ = (m_0c^2 + K_-) + (m_0c^2 + K_+) = K_- + K_+ + 2m_0c^2$$

What is the absolute minimum energy needed to create an electron and a positron?

The minimum energy required in order to create a pair (electron-positron)

$$h\nu = 2m_ec^2 = 1.022\ 10^6\ eV$$

To create an electron-positron pair requires a photon with $\lambda_{max} = 0.012$ Å (gamma rays)

Summing up, we may conclude that the interaction of the photons with matter takes place by means of the photoelectric effect at low energies, the Compton effect at intermediate energies and pair production at high energies.

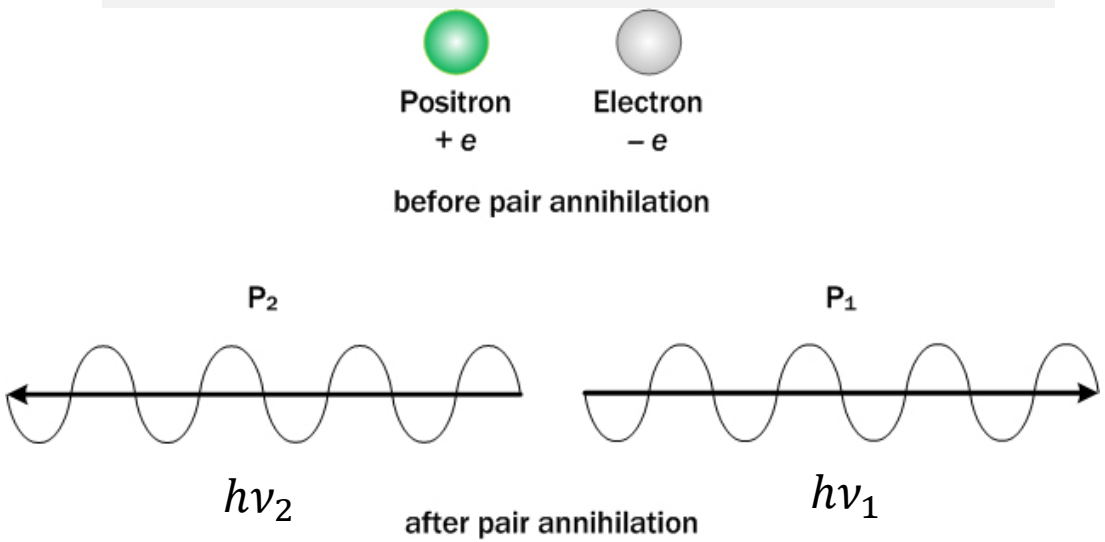
Pair production and Pair annihilation

Pair annihilation

An electron and a positron unite and are annihilated.

Matter disappears and we get radiant energy.

If we assume that the electron and positron have NO initial kinetic energy



Momentum conservation

$$\vec{p}_1 + \vec{p}_2 = 0 \implies |\vec{p}_1| = |\vec{p}_2| \implies \nu_1 = \nu_2 \implies E_1 = E_2$$

Energy conservation

$$m_0c^2 + m_0c^2 = h\nu_1 + h\nu_2 \equiv 2h\nu$$

$$h\nu = m_ec^2 = 0.51\ MeV \implies \lambda = 0.024\ \text{Å}$$

(electron-positron)

Cross Sections for Photon Absorption and Scattering

We now know several photon–matter interactions. Which one actually occurs?

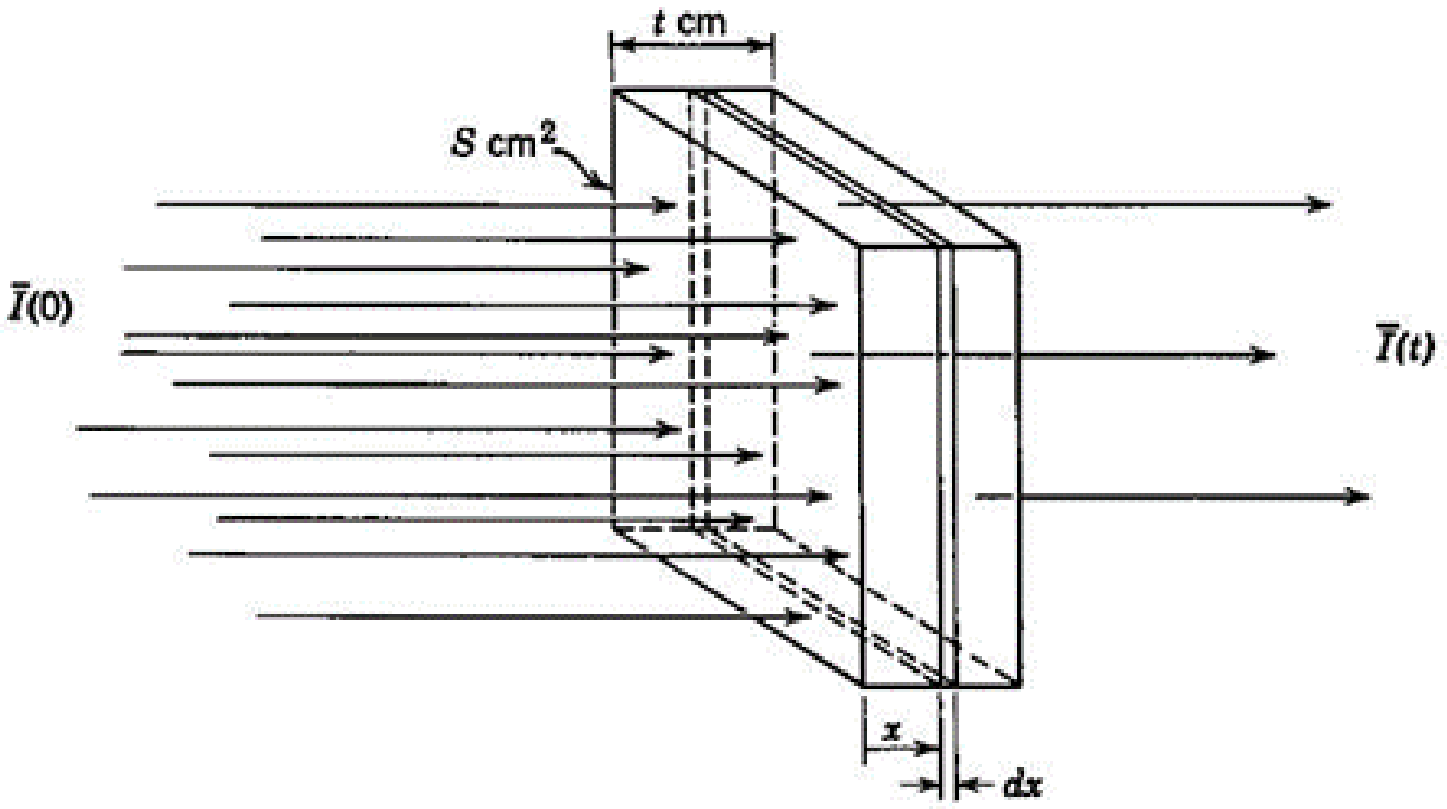


Figure 14–21. A beam of X-rays passing through a foil.

Cross Sections for Photon Absorption and Scattering

A beam of photons passing through a slab of matter. Different processes: photoelectric, pair production, Thompson and Compton scattering.

What is the probability of these processes?

CROSS SECTION

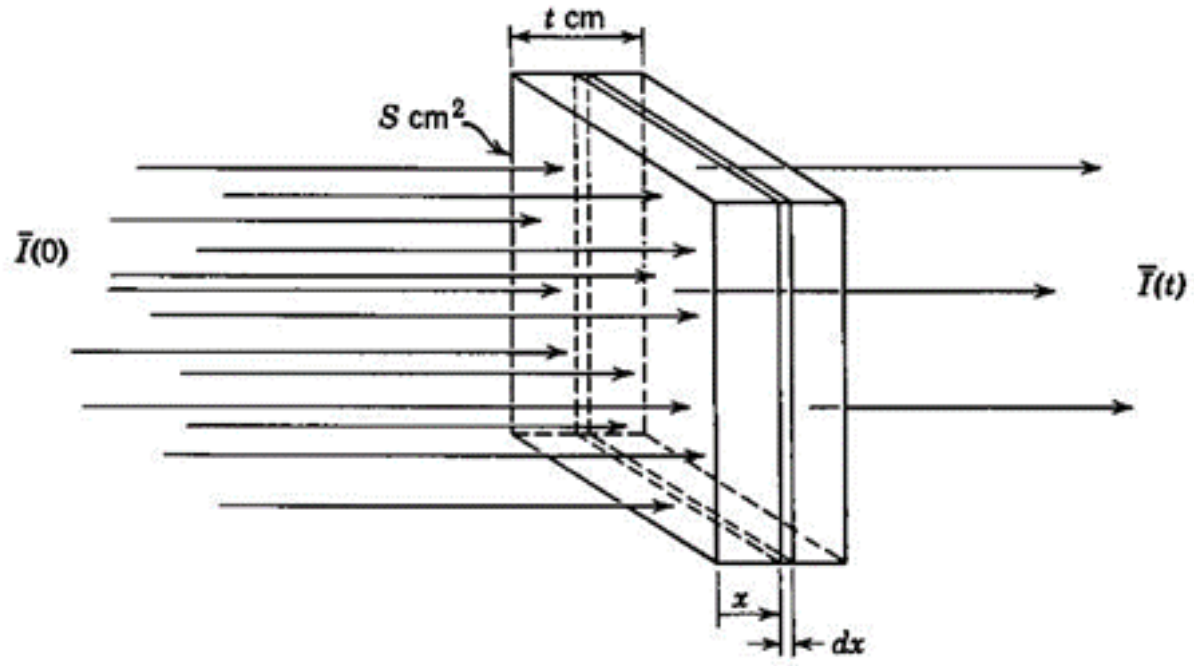


Figure 14-21. A beam of X-rays passing through a foil.

Cross Sections for Photon Absorption and Scattering

Photoelectric cross section

Probability that a photon of a given energy is absorbed in a photoelectric process when passing across the slab is specified by the photoelectric cross section σ_{PE}

N_{PE} : Number of photoelectric absorptions per unit surface and unit time

I : Number of incident photons per unit surface and unit time

n: Number of atoms per unit surface

Definition:

$$N_{PE} = \sigma_{PE} I n$$

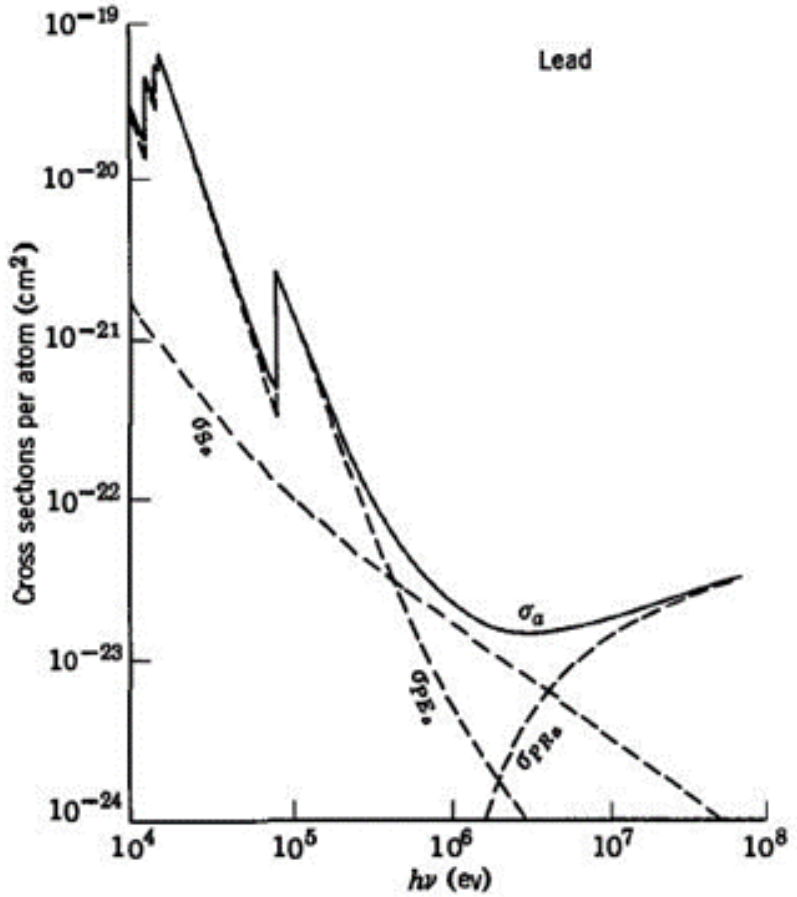
In fact $\sigma_{PE} \Longrightarrow \sigma_{PE}(E_{photon}, material)$

Dimensions of area. The most usual unit is the barn

barn = 10^{-28} m^2 ≡ 10^{-24} cm^2.

Cross Sections for Photon Absorption and Scattering

A cross section is a way of expressing the probability that a certain type of atom will cause a photon of a given energy to undergo a particular process.



14–20. The scattering, photoelectric, pair, and total cross sections per atom

For Pb

Photoelectric effect: $h\nu < 5 \times 10^5$ ev

Scattering: 5×10^5 ev $< h\nu < 5 \times 10^6$ ev

Pair production: 5×10^6 ev $< h\nu$

Because of the different Z dependencies of the cross sections for these processes, the energy ranges in which they dominate σ_a are quite different for atoms of low atomic number. For ^{13}Al these energy ranges are approximately

Photoelectric effect: $h\nu < 5 \times 10^4$ ev

Scattering: 5×10^4 ev $< h\nu < 1 \times 10^7$ ev

Pair production: 1×10^7 ev $< h\nu$